Saturated Superfluid Vacuum VI

Galactic Dynamics: Rotation Curves and Spiral Structure from the Two-Term Source

Stig Norland
Independent Researcher, Bergen, Norway

Draft

Abstract

This paper replaces the former Papers VI-a (*Galactic Standing Waves and Flat Rotation Curves*) and VI-b (*Galactic Morphology as Overtone Structure*). Both were organised around mechanisms driven by the central black hole (CBH): a CBH-anchored standing wave was to produce the flat rotation curve, and CBH spin overtones were to produce the spiral arms. Both mechanisms are falsified (issues #122, #124; falsification record in Appendix A), and one mechanism now covers both phenomena, so one paper covers both subjects.

The surviving architecture has three parts. (i) Gravity is the gradient of the time-dilation field (Paper IV): the rotation curve is $v^2(r) = r\, d\Phi/dr$, with the Keplerian limit near a central mass recovered structurally. (ii) The source of Φ is the *intrinsic two-term structure* of topological defects, measured in the LogSE medium (issue #119, H7a/H8): a concentrated healing-length core and an isothermal $\ell^2\rho/2r^2$ circulation halo, whose Poisson integral is a logarithmic potential $\Phi_\mathrm{h} = v_h^2 \ln r$ — a flat-curve tail with no dark-matter particles. (iii) Everything else is *real Newtonian self-gravity* of the baryons: in pre-registered N-body experiments (200 000 stars, FFT particle-mesh, no black hole, no dark matter), the baryons-only disc reproduces the missing-mass problem (declining curve), the same disc plus the measured halo term gives a flat curve, and spiral arms and a bar grow by swing amplification with no driver — the Ostriker–Peebles instability — directly superseding the CBH-overtone mechanism. The combination of a flat curve *and* growing arms constrains the halo amplitude to an intermediate, maximal-disc balance — a falsifiable condition, not a free parameter — though the halo amplitude's value is not yet derived from the medium (the circulation-halo route to its normalisation was falsified in issue #119, H9).

The data comparison stands on M31 and on the SPARC sample (83 galaxies, pre-registered battery): the *pure* constant-v_h^2 form — extrapolated through the core region the H7a/H8 measurement never covered — is falsified there (not competitive with NFW; 21% inner-quintile overshoot), while the halo amplitude across the population scales as $v_h \propto M_\mathrm{bar}^{0.256\pm0.019}$ with 0.14 dex scatter — the BTFR-equivalent exponent 1/4, parameter-free, now the measured target for any entrainment derivation. On M31, with the measured baryons fixed, the flat plateau requires an additional component supplying $\sim$83% of v^2 whose amplitude is fitted, not predicted. The spiral-structure section records that the pattern's pitch angle is a winding (shear) observable: the pre-registered pitch–Q trend of the old dispersion analysis failed its numerical test (D4b). All retired claims are kept as an explicit falsification record per the programme's negative-results rule.

Contents

1 Introduction

The rotation curves of spiral galaxies stay flat far beyond the radius where the visible matter would let them fall [1], and the standard resolution is a dark-matter halo [2]. The SSV programme's galactic sector originally proposed a different engine: the central black hole. Paper VI-a modelled the disc as a planar soliton phase-locked to a standing wave at the CBH eigenfrequency, producing a Mestel surface density [3] and hence a flat curve; Paper VI-b read the Hubble sequence as the overtone spectrum of the same wave, with CBH spin exciting azimuthal modes $m = 2, 3, 4$.

Both CBH mechanisms are now falsified, each by pre-registered computation against its own decision rules:

- **The radial mechanism** (flat curve from the CBH wave): rebuilding the rotation curve in the correct framework — gravity as the time-dilation-field gradient — showed that the CBH frequency sets at most a wiggle *spacing*; the flat plateau is a free amplitude that must be fitted to each galaxy, not predicted from $M_{\rm BH}$ (issue #122; §6, Appendix A).
- **The azimuthal mechanism** (arms from CBH overtones): it is contradicted observationally — M33 has prominent arms and essentially no central black hole [4], the predicted Milky Way corotation radius misses by more than a factor of 20, and the mode amplitudes invalidly absorbed an astronomically small multipole suppression into their normalisation — and it is superseded numerically: a self-gravitating stellar disc grows a bar and trailing arms with *no* black hole present at all (issue #124; §5, Appendix A).

What replaces them is simpler and uses fewer ingredients. The SSV contribution to galactic dynamics reduces to a single, measured object: the *intrinsic two-term source* carried by topological defects of the medium (Paper IV; issue #119, H7a/H8). Its first term is a concentrated core — the Newtonian monopole. Its second term is the energy of the defect's quantised circulation, an isothermal $1/r^2$ energy density whose Poisson integral is a logarithmic potential — a flat-curve tail. Everything else in a galaxy is ordinary Newtonian self-gravity of the baryons, which we compute with real N-body experiments rather than ansatz potentials. The central black hole appears nowhere in the mechanism; at most it is an inner-boundary perturber.

The paper is organised as follows. Section 2 states the time-dilation-gradient framework and the rotation-curve decomposition. Section 3 imports the measured two-term source. Section 4 presents the pre-registered real-gravity N-body battery (D1–D3) and the maximal-disc balance condition. Section 5 treats spiral structure: swing amplification, the bar, and the pitch angle as a winding observable, including the negative D4b result. Section 6 gives the honest M31 comparison. Section 8 collects claim statuses, open problems, and falsifiable predictions. Appendix A is the falsification record of the retired VI-a/VI-b mechanisms (rule 1 of the programme: correctly negative results are the proof; they are recorded, not deleted). The full retired texts remain available in the repository history.

2 Framework: the Rotation Curve as a Time-Dilation Gradient

Paper IV develops gravity as the gradient of the time-dilation field: the local update availability $A = d\tau/dt = 1 + \Phi/c^2$ slows clocks and bends trajectories, and at leading order all weak-field phenomenology follows from the single potential Φ. For a cold circular orbit at radius r in an axisymmetric galaxy,

$$v^2(r) \;=\; r\,\frac{d\Phi}{dr}, \tag{1}$$

and the potential is the sum of the Poisson integrals of the actual sources present:

$$v^2(r) \;=\; v^2_{\rm bulge}(r) + v^2_{\rm disc}(r) + v^2_{\rm BH}(r) + v^2_{\rm halo}(r). \tag{2}$$

Three structural points, established in the issue-#122 rebuild (`rotation_curve_timedilation`, with tests):

- **The Keplerian limit is recovered, not bolted on.** Near a dominant central mass, $\Phi \to -GM/r$ and (1) gives $v\sqrt{r} = \sqrt{GM}$ constant — the weak-field limit of the time-dilation field, verified in the test suite.
- **Each term is a Poisson integral of a real source.** The bulge and disc terms come from the measured baryonic distributions; the BH term is GM_{BH}/r; the halo term is the Poisson integral of the intrinsic circulation-energy density of §3.
- **No standing-wave amplitude appears.** The former VI-a wave term survives only as a falsified candidate for the halo slot (Appendix A): its amplitude was a free dark-matter-equivalent, and its frequency content set only a wiggle spacing.

3 The Two-Term Source

The SSV-specific input is a measured property of topological defects in the logarithmic-Schrödinger medium (Paper I ontology; Paper IV gravity). In pre-registered relaxation experiments (issue #119, H7a and the H8 cementing battery), an isolated quantised vortex of winding ℓ carries:

1. **a concentrated core** on the healing length — under the Poisson reading $\nabla^2\Phi = 4\pi G\, e/c^2$ this is the Newtonian monopole of the defect's mass-energy;
2. **an isothermal circulation halo**: the kinetic energy density of the quantised circulation,
$$e(r) \;=\; \frac{\ell^2 \rho}{2\, r^2}, \tag{3}$$
measured as $e\cdot r^2 = 0.512 \pm 0.011$ against the parameter-free prediction $\ell^2/2 = 0.5$ at $\ell = 1$ (2.4%), and 2.28 ± 0.13 against 2.0 at $\ell = 2$ — the ℓ^2 law;
3. **its own time-dilation field**: a Bernoulli density depression $\delta\rho \cdot r^2 \to -1/(2b)$, confirmed at two values of the LogSE coupling (0.6% at $b = 0.5$; 11% at $b = 1$).

No radiation, no external drive, and no new postulate enter: the halo is the defect's *own* circulation energy. As a Poisson source, $e \propto 1/r^2$ integrates to

$$\Phi_{\mathrm{h}}(r) \;=\; v_h^2 \ln r \qquad \Longrightarrow \qquad v_{\mathrm{halo}}^2 = v_h^2 \;=\; \mathrm{const}, \tag{4}$$

the flat-curve tail, with v_h the halo amplitude.

Open: the halo amplitude is not derived (#119 H9 falsified the natural route)

The pre-registered attempt to fix v_h from the medium — the circulation-halo triangle $v_{\mathrm{flat}}^2 = G\rho_0\Gamma^2/(2\pi c^2)$ with the entrained circulation Γ locked to the baryons' angular momentum (Fall relation) — **failed on both decision rules**: the implied gravitational coupling misses by $\sim$35 orders of magnitude, and the implied baryonic Tully–Fisher slope is 3/2 against the observed ≈ 4 [5] (`h9-circulation-halo-triangle`). In this paper v_h is therefore an *empirical amplitude per galaxy*, constrained — not fixed — by the balance condition of §4.3. Deriving the entrainment law $\Gamma(M)$ (which must deliver $\Gamma \propto M^{1/4}$ at $\sim$17 orders below the baryon circulation, or fail) is owned by the successor issue #129. Dispersion-supported dwarf spheroidals, which barely rotate yet are dark-matter dominated, were the standing discriminator for any circulation-sourced halo; that discriminator has now been run (§7.2), and it falsifies entrainment proportional to the baryons' rotation.

4 Real-Gravity N-Body Experiments (D1–D3)

4.1 Setup

All claims in this section are backed by direct N-body computation with *real* Newtonian self-gravity — softened $1/r^2$ forces from an FFT particle-mesh solver with isolated (zero-padded, Hockney–Eastwood [6]) boundary conditions — not by ansatz potentials (`disc_nbody`; GPU). The disc is exponential (scale length $R_d = 1$, $G = M_{\text{disc}} = 1$), 2×10^5 equal-mass stars, velocities initialised from the *measured* total acceleration with Toomre-Q-controlled radial dispersion [7]. There is **no central black hole** (the M33-like configuration) and there are **no dark-matter particles**. The only SSV ingredient is optional: the halo term (4) applied as the external acceleration $a_h = -v_h^2/r$, switched on or off. Decision rules D1–D4 were posted to issue #124 before running.

4.2 Results

D1–D3: declining curve, flat curve, and arms — all from real gravity (issue #124)

rule	pre-registered criterion	outcome
D1	baryons-only curve declines beyond the disc	**CONFIRMED** (outer/inner 0.88–0.94)
D2	+ measured halo (4): flat at large r	**CONFIRMED** ($v = 0.79$–0.81 over 1–5.5 R_d)
D3	m=2 amplitude $\geq 10\%$ in a few rotations, no BH	**CONFIRMED** (A_2: $0.014 \to 0.118$, $\sim$3 rotations)

The baryons-only disc develops a *bar plus trailing arms* — the Ostriker–Peebles instability [8] — from pure self-gravity, with the $m = 2$ pattern winding from 48° to 22° pitch. Spiral structure is a collective mode of the self-gravitating baryons (swing amplification [9]); no central driver exists in the simulation, so no central driver is needed. A full three-dimensional run (PM on a 128^3 grid, disc with vertical structure) reproduces the same phenomenology with real 3D gravity. Figure: `figures/fig_nbody_real_gravity.png`.

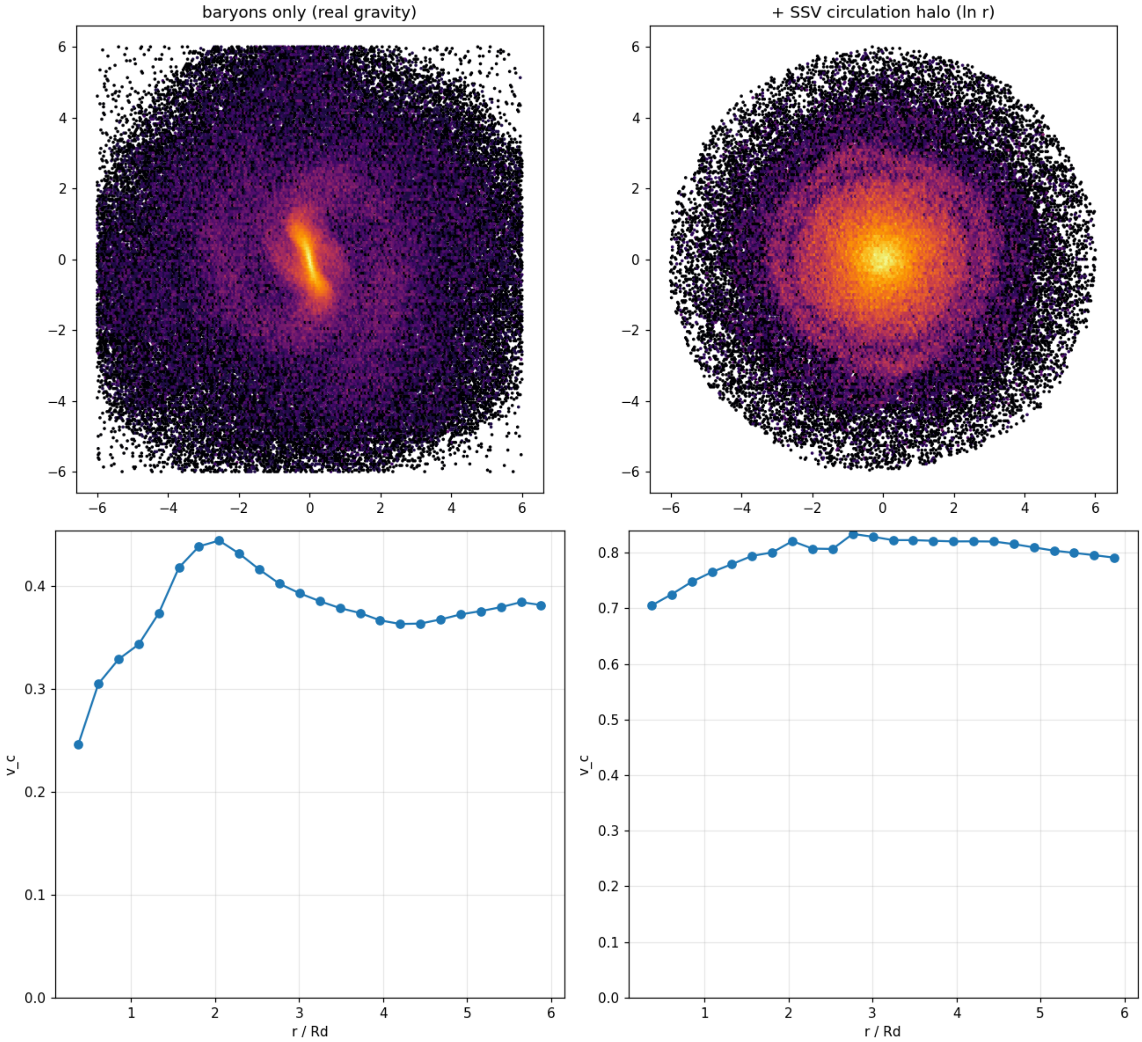

Figure 1: The pre-registered D1–D3 battery. *Left*: baryons-only disc — the rotation curve declines (D1) and a bar + trailing arms grow from pure self-gravity (D3). *Right*: the same disc with the measured SSV circulation halo (4) — the curve is flat (D2). No black hole, no dark-matter particles, real $1/r^2$ forces.

4.3 The maximal-disc balance condition

The honest finding of the battery (rule 1: reported at full strength) is that the halo-dominated configuration *kills the arms*: at $v_h = 0.65$ the disc is stabilised into featurelessness ($A_2 \sim 0.007$) — the known "dead disc" effect of rigid halos. Real galaxies have flat curves *and* spiral arms. In the 3D run the combination is achieved at intermediate amplitude: $v_h = 0.45$ gives a flat curve (outer/inner 1.06–1.10) and growing arms (A_2: $0.004 \rightarrow 0.060$ and rising).

The balance condition

The requirement *flat curve and live spiral structure* forces the disc to remain near-maximal inside with the halo taking over outside. This converts the halo amplitude v_h from a free parameter into a falsifiable balance condition on every spiral galaxy: a galaxy whose photometric disc is far from maximal yet shows strong arms, or a maximal disc with no halo signature in its outer curve, breaks it.

5 Spiral Structure

5.1 Arms without a driver

D3 settles the mechanism question for this programme: spiral arms grow by swing amplification of the self-gravitating baryons, seeded by shot noise, with no external or central driver. The CBH-overtone account of VI-b — arms as spin-excited azimuthal modes of a CBH-anchored wave — is superseded (and independently contradicted by observation; Appendix A). The instability route also produces the bar morphology directly: the baryons-only disc *is* a barred spiral after three rotations.

5.2 Pitch angle is a winding observable, not a dispersion dial (D4/D4b)

The one quantitative VI-b result that survived the mechanism's fall was the marginal-stability pitch relation of its Lin–Shu-type dispersion analysis [10], $\tan\alpha_m = mQ/4$ — pitch increasing with the Toomre parameter. We tested it as pre-registered rule D4: sweep $Q \in \{1.0, 1.3, 1.6, 2.0\}$ at matched times and seeds, measure the $m = 2$ phase-vs-$\ln r$ pitch, demand a monotone increase (Spearman $\rho \geq +0.8$) wherever the arm amplitude is measurable ($A_2 \geq 0.02$).

D4b: the pitch–Q trend FAILS in the real-gravity disc (issue #124)

At the balanced halo amplitude the 2D disc's arms sit below the validity floor (D4: inconclusive by rule). In the strong-arm baryons-only disc (D4b, pre-registered follow-up; all 16 (Q, t) cells valid, $A_2 = 0.047$–0.196):

t	pitch(Q=1.0)	(1.3)	(1.6)	(2.0)	Spearman ρ
20	24.1°	23.5°	45.8°	44.9°	+0.6
30	26.0°	27.5°	27.2°	27.0°	+0.2
40	39.9°	29.7°	27.3°	58.1°	+0.2

No matched time reaches the pre-registered $\rho \geq +0.8$; at $t = 30$ the pitch is *flat to within 1.5° across a factor-2 sweep in Q*. The pitch of the swing-amplified pattern is set by differential-rotation winding (the shear), while Q controls the *amplitude* of the response. The relation $\tan\alpha = mQ/4$ is falsified as a fixed-time prediction in the real-gravity disc (`d4-pitch-vs-q-issue124`; `disc_nbody_d4_pitch`).

What survives of the VI-b analysis: the azimuthal-averaging argument for why 1D rotation curves look smooth (a geometric fact independent of the arm mechanism), and the dispersion relation *as a relation* — linearising any self-gravitating LogSE disc produces a Lin–Shu-type balance — with its pitch corollary now carrying a falsified status.

6 The Honest M31 Comparison

The issue-#122 rebuild fitted the full Chemin et al. [11] H I rotation curve of M31 ($r = 1.14$–38 kpc, $n = 98$, absolute errors) in the time-dilation-gradient framework (`timedilation-rotation-curve-resu`

model	reduced χ^2	RMS (km/s)
SSV time-dilation, all parameters free	10.1	19.8
measured baryons + NFW halo [2]	18.3	30.4
measured baryons fixed + flat-tail component	13.5	28.3

Three honest readings, all recorded in the issue before this paper was assembled:

- None of the models is statistically acceptable on the full range (reduced $\chi^2 = 10$–18): M31's inner bulge/bar and warp make the complete curve genuinely hard for *any* smooth axisymmetric model — this is not SSV-specific.
- The free fit drives the visible disc's contribution toward zero: an unconstrained flat-tail amplitude is a dark-matter-equivalent and will absorb the baryons if allowed.
- With the measured baryons fixed, they supply only $v \approx 110$ km/s of the observed ≈ 266 km/s at 38 kpc: the tail must carry $\sim$83% of v^2, with amplitude ($V \approx 230$ km/s) **fitted, not predicted**. This is the falsification of the old VI-a claim that the CBH wave produces the plateau, and it is exactly the gap the §3 gapbox records: the framework supplies the *form* of the tail ($\ln r$), not yet its amplitude.

7 The SPARC Sample: the Halo at Population Scale

One galaxy is an anecdote. The SPARC database [12] provides 175 galaxies with both the observed rotation curve and the baryonic decomposition ($V_{\rm gas}$, $V_{\rm disk}$, $V_{\rm bul}$) at every radius, and the SSV halo is unusually testable there: because $\Phi_{\rm h} = v_h^2 \ln r$ contributes a *constant* v_h^2 in quadrature, the model

$$v^2_{\rm model}(r) \;=\; V_{\rm gas}|V_{\rm gas}| + \Upsilon_d V^2_{\rm disk} + \Upsilon_b V^2_{\rm bul} + v_h^2, \tag{5}$$

with the SPARC-standard mass-to-light ratios fixed ($\Upsilon_d = 0.5$, $\Upsilon_b = 0.7$), has **one free parameter per galaxy**. The pre-registered battery (issue #133; script `sparc_halo_fit`; primary sample $Q = 1$, $i \geq 30°$, ≥ 10 points: 83 galaxies) fits this against NFW (2 parameters), MOND/RAR (0 parameters, a_0 global), and the endpoint-anchored solid-body model of Flynn & Cannaliato [13] that prompted the exercise. Full numbers: result note `sparc-halo-fit-issue133` and its receipt. **No galaxy is hidden:** all 175 SPARC galaxies are fitted identically and tabulated per galaxy (`sparc-per-galaxy-results`), and the complete rule battery is evaluated on three nested tiers — the pre-registered primary sample, $Q \leq 2$, and *all 175 with no cuts*. Every verdict below is identical at every tier; the rule-(c) slope moves only from 0.256 (primary) to 0.292 ± 0.016 (no cuts), remaining inside the pre-registered window. The cuts are SPARC's own quality recommendations ($Q = 3$ flags major asymmetries; low inclination leaves the velocity scale ill-determined), not a selection made after seeing results.

Rules (a)/(b): the pure (no-core) form is falsified (issue #133)

Median reduced χ^2 over the sample: SSV pure **14.5** versus NFW 1.6 (pre-registered threshold: ratio ≤ 1.5; measured 8.8) — rule (a) FAIL. The stacked residuals show why — rule (b) FAIL with a clean signature: median fractional residuals per quintile of $r/R_{\rm max}$ are -0.21, -0.02, $+0.04$, $+0.08$, $+0.11$. A single constant v_h^2 applied at *all* radii overshoots by 21% in the innermost quintile and consequently undershoots outside. The failure localises exactly where the model exceeded its measured foundation: H7a/H8 established the $1/r^2$ circulation-energy density *outside* the defect core, and the pure form extrapolated it to $r \to 0$. (Context rows: MOND/RAR scores 12.2 with zero per-galaxy parameters; the velocity-addition ω model of [13] scores 93.7 — adding velocities rather than v^2 does not survive error-weighted statistics.)

Rule (c): the halo amplitude scales as $M_{\rm bar}^{0.256\pm0.019}$ — the BTFR-equivalent exponent, parameter-free

Across all 83 galaxies (every fit constrains v_h away from zero):

$$\log_{10} v_h = (0.256 \pm 0.019)\,\log_{10} M_{\rm bar} - 0.665, \qquad \text{scatter } 0.14 \text{ dex}, \tag{6}$$

with $M_{\rm bar} = 0.5\,L_{[3.6]} + 1.33\,M_{\rm HI}$, against the pre-registered BTFR-equivalent prediction of slope 1/4 ($M \propto v^4$ [5]); $v_h(10^{10} M_\odot) \approx 79$ km/s. Robust on the $Q \le 2$ sample (slope 0.268, 115 galaxies). This is the measured curve that any entrainment law $\Gamma(M)$ of issue #129 must reproduce — the empirical target the H9 falsification left undefined. Per rule 1 a passing rule is suggestive, not proof.

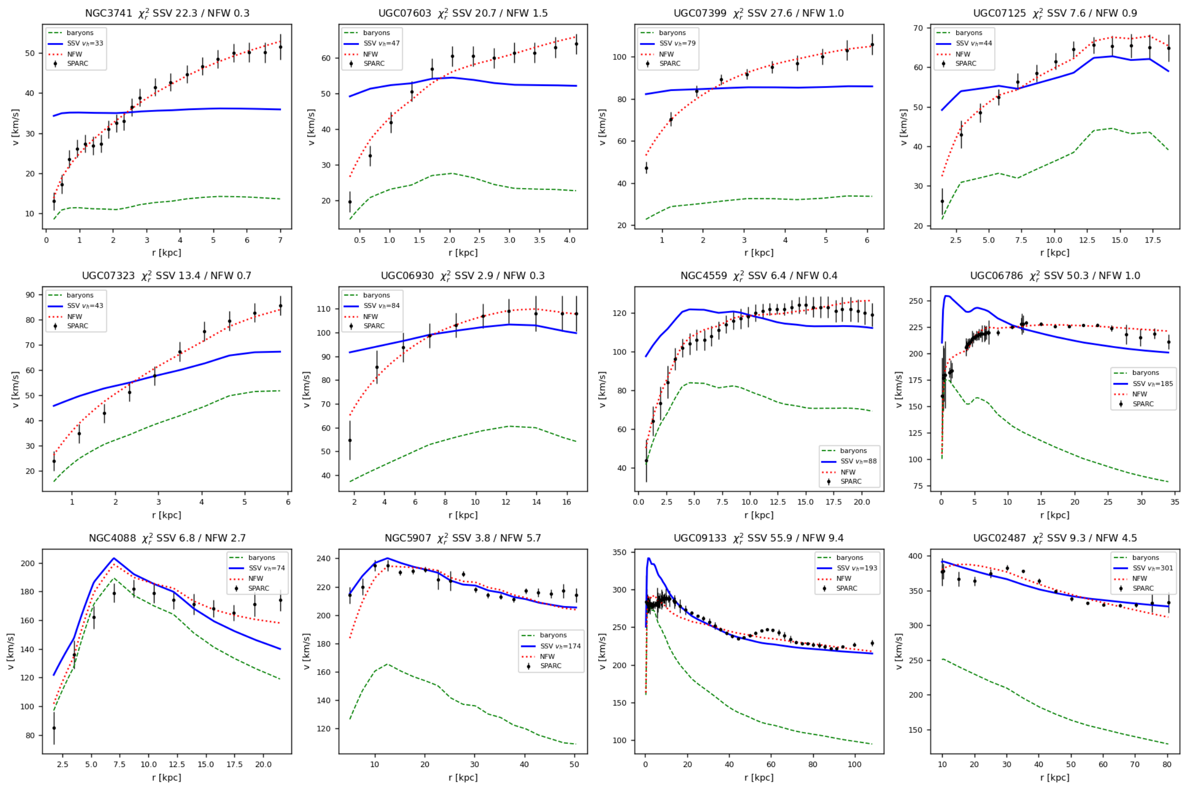

Figure 2: Twelve SPARC galaxies spanning the luminosity range of the $Q = 1$ sample: data (points), fixed baryons (dashed), the pre-registered pure SSV halo (solid blue), and NFW (dotted). The inner-region overshoot of the pure form — the rule-(b) falsifier — is visible in the low-mass discs.

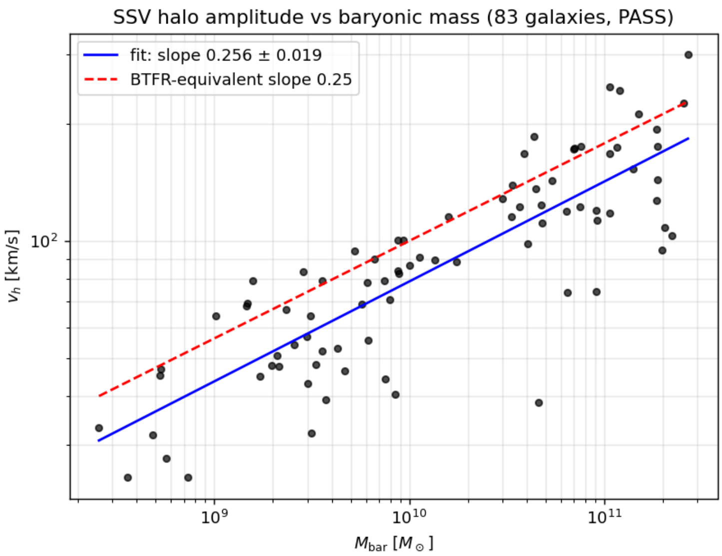

Figure 3: The fitted halo amplitude against baryonic mass for the 83 $Q = 1$ galaxies. The free fit (solid) gives slope 0.256 ± 0.019 with 0.14 dex scatter; the parameter-free BTFR-equivalent slope 1/4 (dashed) is shown for comparison.

Post-hoc exploration (flagged, not pre-registered). Rolling the halo off below a core radius, $v_h^2\, r^2/(r^2 + r_c^2)$, with (v_h, r_c) free, gives median reduced $\chi^2 = 0.89$ and **beats NFW by BIC in 76% of the sample at equal parameter count** (r_c quartiles 2.5/4.3/7.0 kpc) — the entire pre-registered failure is the core region. This is recorded as an exploration only: promoting the cored form requires deriving the core scale from the medium and pre-registering an r_c(galaxy) relation first. That derivation remains open; the dwarf-spheroidal discriminator (dSphs are not in SPARC's rotation-supported sample) has now been run, next.

7.1 The acceleration scale and the cosmological-horizon over-constraint

The constant-v_h halo coincides with the deep-MOND tail with $a_{0,h} \equiv v_h^4/(GM_{\rm bar})$ (issue #146; script `a0_from_vh`); the cored primary-tier value is $a_0 = 1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ (scatter ~ 0.31 dex).

$a_0 = cH_0/2\pi$ from the de Sitter horizon, and the duality over-constraint (H-A0-IR, #155)

Papers V's local horizons fix the holographic thermodynamics — the Unruh/Gibbons–Hawking factor 2π in $kT_H = \hbar g_H/2\pi c$ (S1) and the area-law $\frac{1}{4}$ (S2). Applied to the *cosmological* (de Sitter / Hubble) horizon, the same construction gives the IR acceleration scale

$$a_0 = c\, f_{\rm dS} = \frac{cH_0}{2\pi}, \qquad f_{\rm dS} = \frac{H_0}{2\pi} \text{ (the GH thermal frequency).}$$

This is not a coincidence of coefficients but the *same computation*: the identical finite-difference surface gravity used for Paper V's local dumb-hole, applied to the Hubble flow $v = H_0 r$ against c, returns $g_{\rm dS} = \frac{1}{2}|\partial_r(c^2 - v^2)|_{r_{\rm dS}} = cH_0$ (to 10^{-12}) and hence $a_0 = g_{\rm dS}/2\pi$. The *magnitude* is conceded: H_0 (via $\Lambda = (8\pi G/c^2)(P_0/\rho_0 c^2)$ and the undetermined saturation pressure P_0, Paper VIII) is an input, just as G's magnitude is (#154); only the dimensionless coefficient is tested. The battery `a0_overconstraint` confronts the measured a_0 against each candidate (ratio measured/predicted, $H_0 = 73$): $cH_0/2\pi$ gives 0.999 (and 1.082 at $H_0 = 67.4$ — consistent across the H_0 tension); $c^2\sqrt{\Lambda}$ gives 0.12 — **excluded at** 8.4× (and excluded under the independent RAR scale too),

a clean negative for the cosmological-constant branch. The over-constraint is therefore *survived*: the local screens (V) and the cosmological screen carry one thermodynamics; the otherwise-falsifier-free duality *could* have been falsified here (an a_0 on the $\sqrt{\Lambda}$ form would have done it) and was not.
Survived, not confirmed — and $1/2\pi$ is not singled out (rule 1). Verlinde's $cH_0/6$ differs from $cH_0/2\pi$ by only $6/2\pi = 0.955$ (0.020 dex), far inside the ~ 0.31 dex SPARC scatter (0.065σ). Worse for a clean claim: the *independent, tighter* radial-acceleration-relation scale $a_0 = 1.20 \times 10^{-10}$ [14] mildly favours 1/6 at $H_0 = 73$ (ratio 1.015) while SSV's own halo a_0 favours $1/2\pi$ (ratio 0.999) — the two best estimators straddle the coefficients (∼6% apart), and even the RAR's small random error is blocked by its ~ 0.08 dex systematic and the 0.035 dex H_0 tension. So $1/2\pi$ is SSV's prediction and is consistent, but *not* singled out by the data. The decisive separator is the redshift-evolution falsifier (re-pinned from #146, rule A4): the cH branch predicts a BTFR-normalisation shift $\Delta \log M = \log_{10}[H(z)/H_0] = 0.121/0.253/0.482$ dex at $z = 0.5/1/2$ (flat ΛCDM), against 0 for the $\sqrt{\Lambda}$ branch — a test that needs high-z rotation curves not yet in hand. Result note `h-a0-ir-issue155`.

7.2 The dwarf-spheroidal ledger: the halo reads mass, not rotation

The eight classical Milky Way dwarf spheroidals barely rotate ($v_{\rm rot} \lesssim 3$ km/s; dispersion-supported) yet are the most dark-matter-dominated systems known — the pre-registered battery of issue #147 (script `dsph_ledger`; McConnachie-2012 values, Wolf estimator $v_c(r_{1/2}) = \sqrt{3}\,\sigma_{\rm los}$) confronts them with two competing sourcings of the halo amplitude: the measured mass law of Eq. (6) (model A), and rotation-proportional entrainment $v_h \propto \Gamma_{\rm bar} = 2\pi R\, v_{\rm rot}$ normalized at the H9 Milky-Way reference (model B) — the reading under which "circulation halo" is fed by what the baryons are doing.

#147 B1: rotation-proportional entrainment is falsified at the dwarf spheroidals

Extrapolated *three decades in mass* below the SPARC sample, the dwarfs sit on the mass law: median offset +0.20 dex (no dwarf outside the pre-registered ±0.5 window; the largest offset is +0.43). The rotation law's *upper limits* miss by a median of 2.32 **dex** — a factor ~ 209. The halo amplitude reads baryonic *mass*, not baryonic *rotation*. Three honesty items are recorded with it.
(i) The verdict is sweep-fragile, and the fragility is reported rather than absorbed. Across the 81-point pre-registered sweep (M_*/L varied $\times/\boldsymbol{\nabla} \cdot 2$; $v_{\rm rot}$ limits 1–10 km/s; ±30% anisotropy systematic; and, since #203, the Milky-Way normalisation radius over 4–15 kpc) 6 points return *inconclusive* rather than *falsified*. All 6 sit at $v_{\rm rot} = 10$ km/s — more than three times the ≤ 3 km/s observational upper limit the sweep exists to stress — and at the smallest, most model-B-favourable normalisation radius. Restricted to $v_{\rm rot} \leq 3$ km/s the verdict holds at every sweep point, with 0.17 dex of margin above the 1.5 dex threshold. B1 is therefore falsified throughout the observationally allowed region and inconclusive only outside it; the unqualified stability flag in the receipt is `false`.
(ii) The naive "no rotation ⇒ no circulation budget" killer does *not* bind — even at $v_{\rm rot} = 1$ km/s the dwarfs carry $\gtrsim 1.4 \times 10^{15}$ times the H9-inverted circulation requirement, so it is the *correlation*, not the budget, that discriminates.
(iii) The dwarfs sit systematically *above* the relation by +0.20 dex (they are MW satellites; the tidal confound is recorded, and at face value the offset means slightly *more* halo than the mass law predicts, in the same direction the relation's 0.14-dex scatter allows).

#147 B3: a universal kpc-scale core is falsified at dwarf scales

Under the post-hoc cored form with a *universal* $r_c = 2.5$ kpc (the SPARC lower quartile — the choice most favorable to dwarf halos), the predicted dispersions collapse to $\sigma_{\rm pred} \approx 1$–6 km/s against observed 6.6–11.7 km/s: **7 of 8 dwarfs fall below half the observed dispersion** (threshold: ≥ 6 of 8), sweep-stable. The #133 promotion condition sharpens accordingly: any r_c(galaxy) relation must deliver $r_c \lesssim r_{1/2} \sim 0.2$–0.9 kpc at dwarf scale — a single medium-set core length is dead; only a baryon-set core scale survives.

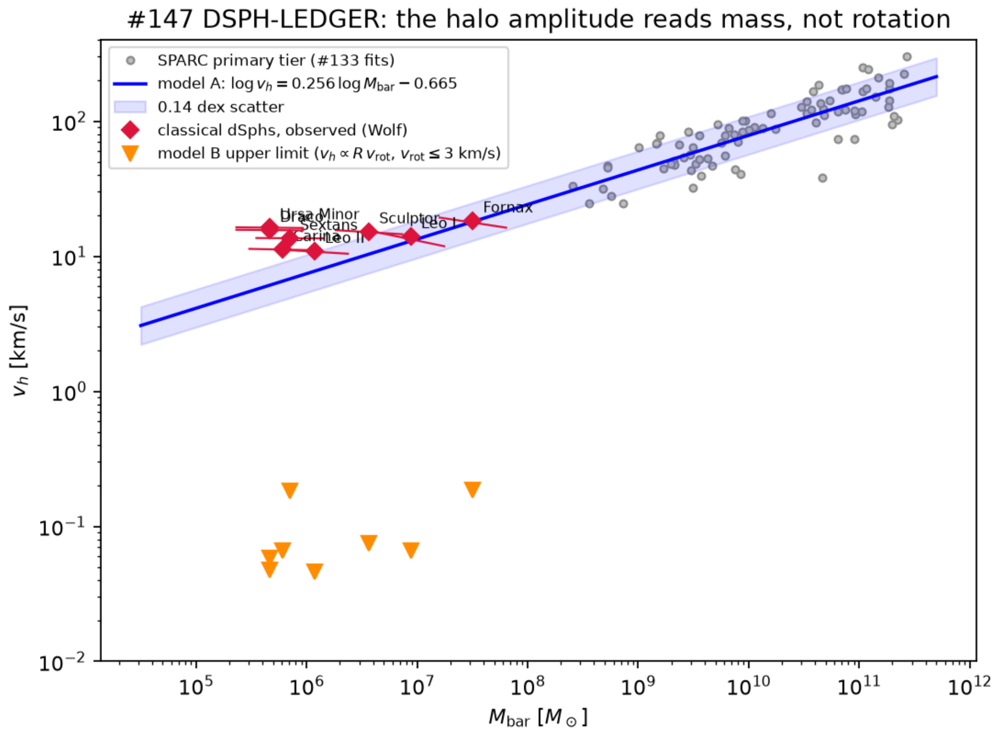

Figure 4: The dwarf-spheroidal ledger (#147). The eight classical dSphs (diamonds; bars span the M_*/L sweep) against the SPARC primary tier (grey) and the measured mass law Eq. (6) (line, with 0.14 dex band). Triangles: the rotation-proportional entrainment *upper limits* at $v_{\rm rot} \leq 3$ km/s — short by ~ 2.32 dex. The halo amplitude tracks mass, not rotation. The triangles sit two decades below the data because classical dSphs genuinely do not rotate: matching the observed v_h under a law linear in $\Gamma_{\rm bar}$ would require rotation speeds of 224–1030 km/s in systems whose *entire* velocity dispersion is $\sigma_{\rm los} \approx 7$–12 km/s — every one of them also above the Milky Way's own rotation.

Two limitations of the ledger, recorded. *Why model B is linear in* Γ. The exponent is not a free choice: model A gives $v_h \propto M_{\rm bar}^{0.256}$ and H9 gives $\Gamma_{\rm req} \propto M_{\rm bar}^{1/4}$, so $v_h \propto \Gamma^{1.02}$. The alternative readings are worth stating because they are self-defeating rather than merely disfavoured: $v_h \propto \Gamma^{1/4}$ would lift model B onto model A (median offset -0.03 dex), so the two hypotheses B1 is pre-registered to separate would become the same curve and the test would decide nothing. An exponent chosen to bring the triangles of Fig. 4 up to the data is an exponent that dissolves the discriminator.

The mass estimator assumes Newtonian dynamics. The Wolf $v_c(r_{1/2}) = \sqrt{3}\,\sigma_{\rm los}$ relation is a spherical Jeans result in a Newtonian potential. General-relativistic corrections are irrelevant here — $(v_c/c)^2 \lesssim 5 \times 10^{-9}$ against a 0.14 dex observational scatter — but the Newtonian assumption is not: every classical dSph sits at $a = v_c^2/r_{1/2} \approx 0.06$–$0.32\, a_0$, i.e. entirely *below* the acceleration scale that §7.1 argues governs this regime. The ledger therefore extracts v_h with a tool that assumes the dynamics this paper disputes. That does not rescue model B, and the reason is structural rather than numerical: $v_h^2 = v_c^2 - v_{\rm bar}^2$ enters *both* residuals, so any error in the extraction is common-mode. Moving model B onto the data needs ~ 2 dex; the same shift would carry the mass-law offset from $+0.20$ to ~ 2.2 dex and breach the ± 0.5 window by fourfold. No systematic in the v_h extraction can rescue model B without destroying model A. What does survive as an open question is narrower: model A is fitted to rotation-supported discs and applied here to pressure-supported systems three decades lower in mass, so a regime-dependent extraction bias would enter the $+0.20$ dex offset — against which the dwarfs' landing on the relation across that whole extrapolation is evidence, not proof.

The consequence for the programme is structural. What survives of "circulation sourcing" after #147 is only the variant in which the galactic medium circulates *without* tracking the baryons' rotation — entrainment as a collective property coupled to mass-energy. Combined with the #129 solar-system scope limit (the halo must not attach to individual stars), the mechanism space is pinched from both sides; the complementary medium-side question — whether the bare medium's static far field can read mass-energy at all — is the pre-registered extensivity battery of issue #149. The full record lives in `dsph-ledger-issue147` and on issues #129 and #147.

8 Claim Status, Predictions, and Open Problems

8.1 Claim-status table

claim	status
$v^2 = r\,d\Phi/dr$ with Keplerian limit near a point mass	*derived* (Paper IV framework; tested, `rotation_curve_timedilation`)
defect carries core + $\ell^2\rho/2r^2$ circulation halo + Bernoulli depression	*measured* in the LogSE medium (#119 H7a/H8; ℓ^2 law and $-1/(2b)$ tail confirmed)
halo Poisson integral $\Rightarrow$ flat tail $v_h^2 \ln r$	*derived* (exact integration factor pinned by test)
D1 declining baryons-only curve; D2 flat with halo; D3 arms/bar without BH	*numerical*, pre-registered, confirmed (#124)
maximal-disc balance condition on v_h	*numerical constraint* (3D balance run); falsifiable per galaxy
halo amplitude $v_h(M)$ from the medium	*open* — the natural chain is *falsified* (#119 H9: 35 orders + slope 3/2 vs 4); owned by #129
pitch–Q trend ($\tan\alpha = mQ/4$)	*falsified* in the real-gravity disc (D4b)
pure (no-core) constant-v_h^2 halo on SPARC	*falsified* (#133 rules a/b: not competitive; 21% inner-quintile overshoot — the extrapolation past the measured core window)
$v_h \propto M_{\rm bar}^{0.256\pm0.019}$ over 83 galaxies, 0.14 dex	*measured* (#133 rule c PASS vs parameter-free 1/4; suggestive, and the falsification target for #129)
cored halo: median $\chi_r^2 = 0.89$, BIC-beats NFW in 76%	*post hoc, flagged* — promotion requires a derived r_c (#133)
halo sourced by entrained baryonic rotation ($v_h \propto R\,v_{\rm rot}$)	*falsified* (#147 B1: classical dSphs sit on the mass law at +0.20 dex while the rotation law's upper limits miss by 2.32 dex; sweep-fragile — 6/81 points inconclusive, all above the observational $v_{\rm rot}$ limit, #203)
$v_h(M_{\rm bar})$ extends to dispersion-supported dwarfs (3 decades below SPARC)	*measured* (#147: median +0.20 dex; suggestive per rule 1; tidal confound and the Newtonian-estimator assumption recorded)
universal (medium-set) core radius r_c	*falsified at dwarf scales* (#147 B3: 7/8 dwarfs below half the observed σ; surviving r_c(galaxy) relations must be baryon-set, $r_c \lesssim r_{1/2}$)
CBH standing wave produces the flat curve	*falsified* (#122; Appendix A)
CBH overtones produce arms / Hubble sequence	*falsified/superseded* (#124; Appendix A)
$a_0 = cH_0/2\pi$ coefficient from the de Sitter horizon	*derived* (coefficient; over-constraint survived, §7.1, #155) — same Unruh/GH 2π as Paper V's local horizons
a_0 magnitude (H_0/Λ via P_0)	*conceded input* (Paper VIII gapbox; parallel to G, #154)
$a_0 = c^2\sqrt{\Lambda}$ (cosmological-constant form)	*falsified* (#155: 8.4× too large)

8.2 Falsifiable statements this paper makes

1. **Balance:** every spiral with a flat outer curve and live arms must be near-maximal-disc inside; strong arms in a strongly halo-dominated disc break the picture (§4.3).
2. **BH-independence:** arm and bar incidence must not require a central black hole at any mass (M33 is the existence proof; a demonstrated CBH-mass–arm-existence dependence

would break §5).

3. **Pitch:** pitch angle tracks shear/winding age, not the dispersion at fixed time (§5.2).
4. **dSph discriminator (run, §7.2):** rotation-fed circulation sourcing died there (#147 B1). What remains falsifiable: any surviving entrainment law must couple to mass-energy rather than rotation and must keep wide binaries strictly Keplerian (the #129 scope limit), and any r_c(galaxy) relation must shrink to $\lesssim r_{1/2}$ at dwarf scale (B3) — a measured dwarf with a kpc-scale core, or a demonstrated v_h–rotation correlation at fixed $M_{\rm bar}$, breaks this picture.

8.3 Open problems

The halo amplitude's derivation (and with it the baryonic Tully–Fisher normalisation and slope) is the sector's central open problem, owned by issue #129 together with the gravitational coupling's magnitude. The CBH's residual role — whether its inner Keplerian region leaves any observable imprint (wiggle spacing was the last candidate, and its amplitude chain is retired) — is open and currently unsupported. Both are recorded as open, with no claim made.

Two further handles are now pinned (§7.1, #155). The acceleration scale's coefficient $a_0 = cH_0/2\pi$ is consistent with the de Sitter Gibbons–Hawking factor but *not decisively separable* from Verlinde's $cH_0/6$ given the SPARC scatter (0.065σ); the decisive separator is the redshift-evolution falsifier (the cH branch shifts the BTFR normalisation by 0.12–0.48 dex out to $z = 2$, the $\sqrt{\Lambda}$ branch by zero), which awaits high-z rotation curves. And the a_0/Λ *magnitude* reduces to the saturation pressure $P_0(\rho_0)$ (Paper VIII), conceded as an input in the same sense as G.

9 Conclusion

A galaxy, in the surviving SSV picture, is baryons under real gravity plus the intrinsic circulation halo of the vacuum's defect structure. Real gravity does what it has been known to do since the 1970s: it declines where mass runs out, and it amplifies spiral structure from noise. The medium contributes exactly one new term — a measured, isothermal $1/r^2$ energy density whose potential is logarithmic — and that term supplies the flat tail. The central black hole drives nothing. What the framework does not yet supply is the halo amplitude: its natural derivation route was tested and falsified, and per the programme's rules that negative is stated as the result it is. At population scale the picture sharpened in both directions: the pure constant-v_h^2 form fails on SPARC exactly where it extrapolated past its measured foundation (the core region), while the halo amplitude follows $v_h \propto M_{\rm bar}^{0.256\pm0.019}$ with 0.14 dex scatter — the BTFR-equivalent exponent, measured across 83 galaxies. The sector's forward path — the entrainment law that must produce that exponent, the core-scale derivation, the Tully–Fisher normalisation, the dwarf-spheroidal discriminator — is owned by issues #129 and #133.

A Falsification Record of the Retired VI-a/VI-b Mechanisms

Per programme rule 1, the retired mechanisms are recorded here with their falsifiers. The complete retired texts (including the Mestel-soliton variational derivation, the 2D beam-convolved resonance fits, the $\varepsilon_m(a_{\rm BH})$ overtone tables, and the morphology zoo) are preserved in the repository history under `papers/SSV-VI-a/` and `papers/SSV-VI-b/` at commit `cc59d19` and earlier.

A.1 The CBH standing-wave rotation-curve mechanism (VI-a)

Falsified: the flat plateau is not produced by the CBH (issue #122)

The mechanism claimed a disc soliton phase-locked to a CBH-anchored standing wave at $f_{\rm BH} = f_p\,(m_p/M_{\rm BH})$, yielding a Mestel surface density and an exactly flat baryonic curve. Rebuilt in the correct time-dilation-gradient framework with M31's real baryons fixed, the wave component must supply $\sim$83% of v^2 with its amplitude **fitted** ($V_w = 230$ km/s), not predicted from $M_{\rm BH}/f_{\rm BH}$; the CBH frequency sets only the wiggle spacing Δr. A mechanism whose amplitude is free is a dark-matter parametrisation, not a prediction. Carried problems that die with it: the calibration constant $\mathcal{C} = 1.8 \times 10^9\,{\rm kpc}\,M_\odot$ was numerological (anchored to one galaxy), and the wavelength scaling carried an unresolved $\lambda_{\rm BH} \propto M_{\rm BH}$ versus $\Delta r \propto 1/M_{\rm BH}$ tension. The plane-of-satellites reading (corotating dwarfs [15] as a nodal sheet of the same wave) was a corollary of the falsified wave and is retired untested. The Mestel profile itself [3] remains, of course, a correct description of a $\Sigma \propto 1/r$ disc; what is retired is the claim that a CBH-locked LogSE soliton puts the baryons in that profile.

A.2 The CBH-overtone arm mechanism (VI-b)

Falsified and superseded: arms need no black hole (issue #124)

The mechanism read spiral arms as CBH-spin-excited azimuthal modes ($m = 2, 3, 4$) of the VI-a wave, and the Hubble sequence as its overtone spectrum. Four independent falsifiers:

1. **M33**: prominent grand-design-class arms with $M_{\rm BH} \lesssim 1.5 \times 10^3\,M_\odot$ [4] — the predicted node spacing exceeds the size of the observable universe's disc analogue; the mechanism predicts *no* arms.

2. **Corotation**: $r_c = m\mathcal{C}/(\pi M_{\rm BH})$ gives 266 kpc for the Milky Way against the observed 6–12 kpc — off by more than 20×.

3. **Multipole suppression**: the $\varepsilon_m(a_{\rm BH})$ amplitudes absorbed a Hansen-coefficient suppression of order $(r_g/\Delta r)^m \sim (5 \times 10^{-10})^m$ "into the normalisation"; the true spin-driven amplitudes are $\sim 10^{-19}$, not percent-level. The percent-level table (and the pitch-angle–$M_{\rm BH}$ anti-correlation built on it) was unfounded.

4. **Winding**: the pinned-refreshed-mode picture contradicted the pitch-from-winding picture used in the same chapter, and the proposed $J_m(kr)\cos m\phi$ pattern is not a spiral.

Superseded by: §5 — swing amplification of the self-gravitating baryons produces bar + arms with no BH in the simulation at all (D3). The pitch–Q corollary of the surviving dispersion analysis subsequently failed its own pre-registered test (D4b, §5.2).

B Code and Issue References

C Code and Issue References

Each reference below is pinned so it can be checked directly against the repository (`https://github.com/StigNorland/SVT`): issues link to their tracker entry, and each script or report links to the exact version — the commit that last modified it. Issue numbers in the text hyperlink to this list.

Issues.

- #119 — https://github.com/StigNorland/SVT/issues/119
- #122 — https://github.com/StigNorland/SVT/issues/122
- #124 — https://github.com/StigNorland/SVT/issues/124
- #129 — https://github.com/StigNorland/SVT/issues/129
- #133 — https://github.com/StigNorland/SVT/issues/133
- #146 — https://github.com/StigNorland/SVT/issues/146
- #147 — https://github.com/StigNorland/SVT/issues/147
- #149 — https://github.com/StigNorland/SVT/issues/149
- #154 — https://github.com/StigNorland/SVT/issues/154
- #155 — https://github.com/StigNorland/SVT/issues/155
- #203 — https://github.com/StigNorland/SVT/issues/203

Code (each pinned to the commit that last modified it).

- instruments/paper_vi/a0_from_vh.py @ 0042efa — https://github.com/StigNorland/SVT/blob/0042efa/instruments/paper_vi/a0_from_vh.py
- instruments/paper_vi/a0_overconstraint.py @ fb2d69b — https://github.com/StigNorland/SVT/blob/fb2d69b/instruments/paper_vi/a0_overconstraint.py
- instruments/paper_vi/dsph_ledger.py @ 7219a99 — https://github.com/StigNorland/SVT/blob/7219a99/instruments/paper_vi/dsph_ledger.py
- instruments/paper_vi/sparc_halo_fit.py @ 866cf0a — https://github.com/StigNorland/SVT/blob/866cf0a/instruments/paper_vi/sparc_halo_fit.py
- instruments/paper_vi_a/rotation_curve_timedilation.py @ 24271b8 — https://github.com/StigNorland/SVT/blob/24271b8/instruments/paper_vi_a/rotation_curve_timedilation.py
- instruments/paper_vi_b/disc_nbody.py @ 3428626 — https://github.com/StigNorland/SVT/blob/3428626/instruments/paper_vi_b/disc_nbody.py
- instruments/paper_vi_b/disc_nbody_d4_pitch.py @ 24271b8 — https://github.com/StigNorland/SVT/blob/24271b8/instruments/paper_vi_b/disc_nbody_d4_pitch.py

Reports (result notes, pinned to their last commit).

- papers/SSV-IV/results/h9-circulation-halo-triangle.md @ 92cd3f6 — https://github.com/StigNorland/SVT/blob/92cd3f6/papers/SSV-IV/results/h9-circulation-halo-triar md

- papers/SSV-VI/results/d4-pitch-vs-q-issue124.md @ 24271b8 — https://github.com/StigNorland/SVT/blob/24271b8/papers/SSV-VI/results/d4-pitch-vs-q-issue124.md
- papers/SSV-VI/results/dsph-ledger-issue147.md @ 4cd12bd — https://github.com/StigNorland/SVT/blob/4cd12bd/papers/SSV-VI/results/dsph-ledger-issue147.md
- papers/SSV-VI/results/h-a0-ir-issue155.md @ fb2d69b — https://github.com/StigNorland/SVT/blob/fb2d69b/papers/SSV-VI/results/h-a0-ir-issue155.md
- papers/SSV-VI/results/sparc-halo-fit-issue133.md @ 866cf0a — https://github.com/StigNorland/SVT/blob/866cf0a/papers/SSV-VI/results/sparc-halo-fit-issue133.md
- papers/SSV-VI/results/sparc-per-galaxy-results.md @ 866cf0a — https://github.com/StigNorland/SVT/blob/866cf0a/papers/SSV-VI/results/sparc-per-galaxy-results.md
- papers/SSV-VI/results/timedilation-rotation-curve-result.md @ 24271b8 — https://github.com/StigNorland/SVT/blob/24271b8/papers/SSV-VI/results/timedilation-rotation-cur md

Change record. This paper states its *current* status only. Corrections, withdrawals and reframings are recorded in https://github.com/StigNorland/SVT/blob/main/papers/SSV-VI/CHANGELOG.md, and the full history is in the repository's commits.

References

[1] V. C. Rubin, W. K. Ford, and N. Thonnard, "Rotational properties of 21 Sc galaxies with a large range of luminosities and radii," *Astrophys. J.* **238**, 471–487 (1980).

[2] J. F. Navarro, C. S. Frenk, and S. D. M. White, "A universal density profile from hierarchical clustering," *Astrophys. J.* **490**, 493–508 (1997).

[3] L. Mestel, "On the galactic law of rotation," *Mon. Not. R. Astron. Soc.* **126**, 553–575 (1963).

[4] K. Gebhardt et al., "M33: a galaxy with no supermassive black hole," *Astron. J.* **122**, 2469–2476 (2001).

[5] S. S. McGaugh, "The baryonic Tully–Fisher relation of gas-rich galaxies as a test of ΛCDM and MOND," *Astron. J.* **143**, 40 (2012).

[6] R. W. Hockney and J. W. Eastwood, *Computer Simulation Using Particles*, Adam Hilger, Bristol (1988).

[7] A. Toomre, "On the gravitational stability of a disk of stars," *Astrophys. J.* **139**, 1217–1238 (1964).

[8] J. P. Ostriker and P. J. E. Peebles, "A numerical study of the stability of flattened galaxies: or, can cold galaxies survive?" *Astrophys. J.* **186**, 467–480 (1973).

[9] A. Toomre, "What amplifies the spirals?" in *The Structure and Evolution of Normal Galaxies*, eds. S. M. Fall and D. Lynden-Bell (Cambridge University Press), 111–136 (1981).

[10] C. C. Lin and F. H. Shu, "On the spiral structure of disk galaxies," *Astrophys. J.* **140**, 646–655 (1964).

[11] L. Chemin, C. Carignan, and T. Foster, "H I kinematics and dynamics of Messier 31," *Astrophys. J.* **705**, 1395–1415 (2009).

[12] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves," *Astron. J.* **152**, 157 (2016).

[13] D. C. Flynn and J. Cannaliato, "A new empirical fit to galaxy rotation curves," arXiv:2601.00522 (2026).

[14] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial acceleration relation in rotationally supported galaxies," *Phys. Rev. Lett.* **117**, 201101 (2016).

[15] R. A. Ibata et al., "A vast, thin plane of corotating dwarf galaxies orbiting the Andromeda galaxy," *Nature* **493**, 62–65 (2013).